AGENTIC AI-BASED DYNAMIC TARIFF OPTIMIZATION FOR EV CHARGING NETWORKS USING LARGE-SCALE CHARGING SESSION DATA

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Executive Summary

- Built a three-agent system — demand forecasting → dynamic pricing → monitoring & learning — on 2.1M zone-hour records (Shenzhen) and ~15k charging sessions (US).
- Demand model forecasts utilization at $R^2=0.95$; the pricing agent flattens the demand curve at near revenue-neutrality (-0.4%).
- Core insight: in this low-utilization, price-inelastic fleet network, dynamic pricing's value is demand-balancing and congestion relief — not revenue growth.
- The learning loop closes the feedback cycle, selecting the reward-optimal pricing intensity (0.6) and confirming that conservative, consumer-friendly pricing is also the optimal choice.
- Outcome: ~54% relative reduction in congestion and a positive off-peak demand shift, achieved at near-zero revenue cost.

Data Landscape & Preprocessing

UrbanEV:

- 247 zones × 8,640 5-min intervals → 2.13M records
- Volume, occupancy, price, duration matrices
- Primary modeling data — real price variation

ACN:

- 14,999 charging sessions
- Used for revenue/pricing evaluation
- Separate network — never merged

Preprocessing:

- Wide→long reshape; zero missing values
- Occupancy capped (1 zone); 60% cost assumption
- 5 features: utilization, revenue/session, energy cost, queue proxy, occupancy density

Key EDA Findings

A predominantly under-utilized, overnight-peaked fleet network.

- 61% of intervals below 30% utilization; only 1% above 80% — under-utilized, not congested
- Inverted demand cycle: peaks overnight (~0.33), troughs midday (~0.23) — fleet signature, not commuters
- Near-identical 7 days a week — no weekend drop
- Price elasticity concentrated in non-CBD / fast-charger zones (-0.30); CBD zones unresponsive (-0.05)

Visuals — hourly demand curve and utilization band split — go in the appendix.

Demand Prediction Agent

Accurate, leakage-free, honestly benchmarked forecasting.

- Three outputs: utilization (XGBoost, R^2 0.95), congestion probability (ROC-AUC 0.997), charging load (R^2 0.97)
- 28% better RMSE than naive persistence — captures real demand dynamics, not just autocorrelation
- Leakage-free lag/rolling features; chronological split (no shuffling, since demand trends upward)
- Demand is highly autocorrelated: previous-hour + 3-hour rolling mean drive 87% of predictive power

Visuals → appendix: the 4-model comparison table (naive/linear/RF/XGB) and the feature-importance bar chart.

Dynamic Tariff Optimization

Network-aware pricing that surcharges busy hours, discounts quiet ones

- Two strategies compared: rule-based threshold vs optimised smooth — smooth wins at every elasticity level
- Critical fix: re-centred pricing on the network's real mean utilization (0.288), not a generic 50% — otherwise every hour got discounted
- Near revenue-neutral (-0.4%): prices range ₹14.5 midday → ₹15.5 overnight, mirroring demand
- Revenue insensitive to price bounds (±10–50%) — no revenue case for drastic, consumer-unfriendly swings

Visuals → appendix: the hourly price profile (overnight high / midday low) and the elasticity-sweep table.

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Monitoring & Learning Agent

The feedback loop closes — and confirms conservative pricing is optimal

- Congestion reduced 54% (relative); +0.45% off-peak demand uplift; pricing efficiency near-neutral (-0.56%)
- Learning loop scans pricing intensities → reward peaks at strength 0.6 (a robust plateau, not a knife-edge)
- Beyond 0.6, extra intensity only sacrifices revenue with no further congestion benefit
- 0.6 is fed back as the deployed pricing setting — closing the monitoring → pricing feedback loop

Visual → appendix: the reward-landscape curve (reward vs pricing strength, peak marked at 0.6)

Peak-Shaving & Consolidated Results

The system flattens the demand curve at near-zero revenue cost

- Before vs after pricing: peak-hour utilization -2.8%, off-peak lifted — a clear peak-shaving effect
- Peak suppressed more than the average → demand curve flattens, not shrinks
- All three agents, all brief metrics, consolidated in one scorecard
- Net result: ~54% congestion relief + off-peak shift, achieved at -0.4% revenue (essentially neutral)

Visual → appendix: the before/after utilization bar chart and the consolidated scorecard table.

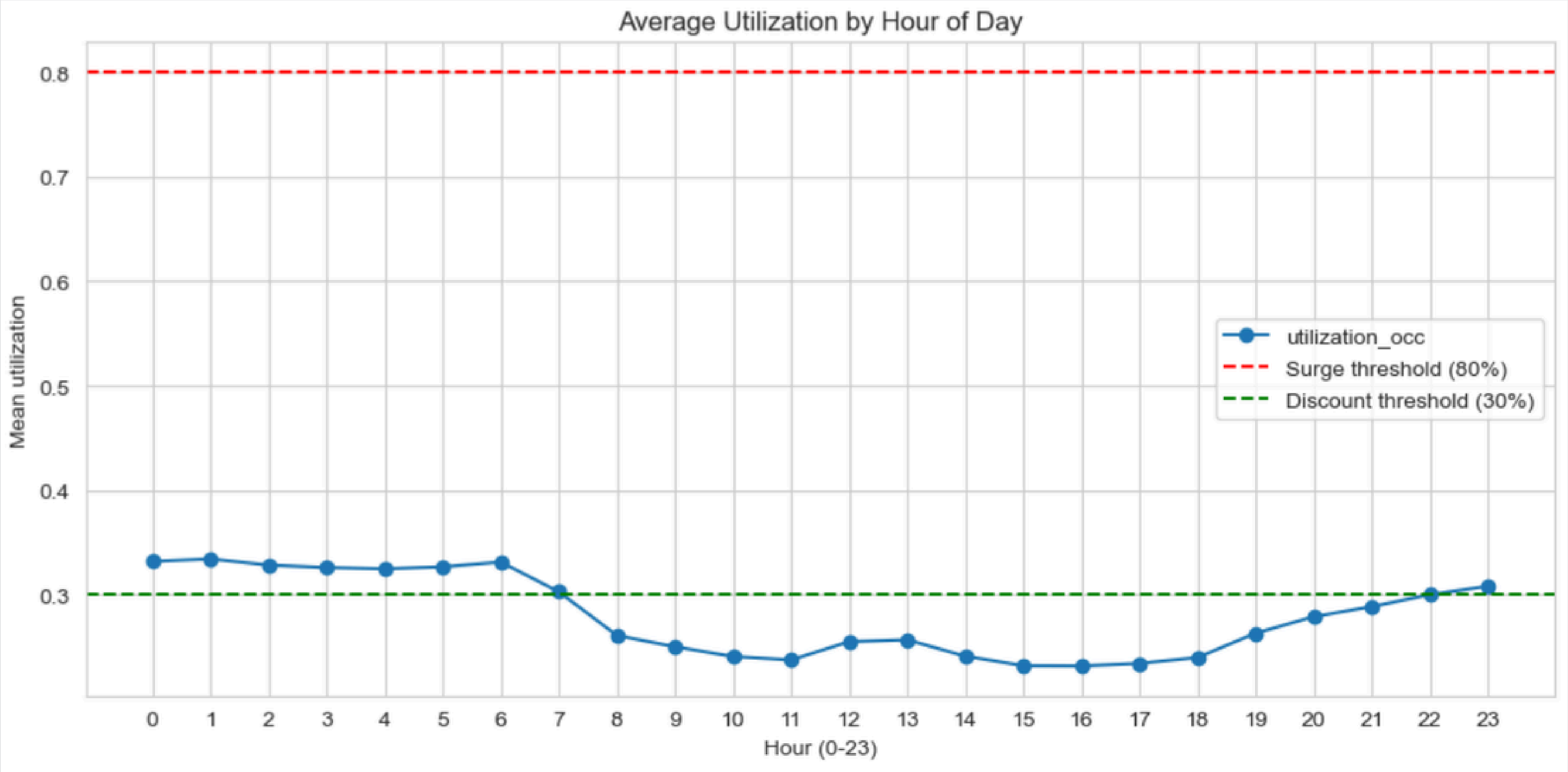
Business, Operational & Policy Implications

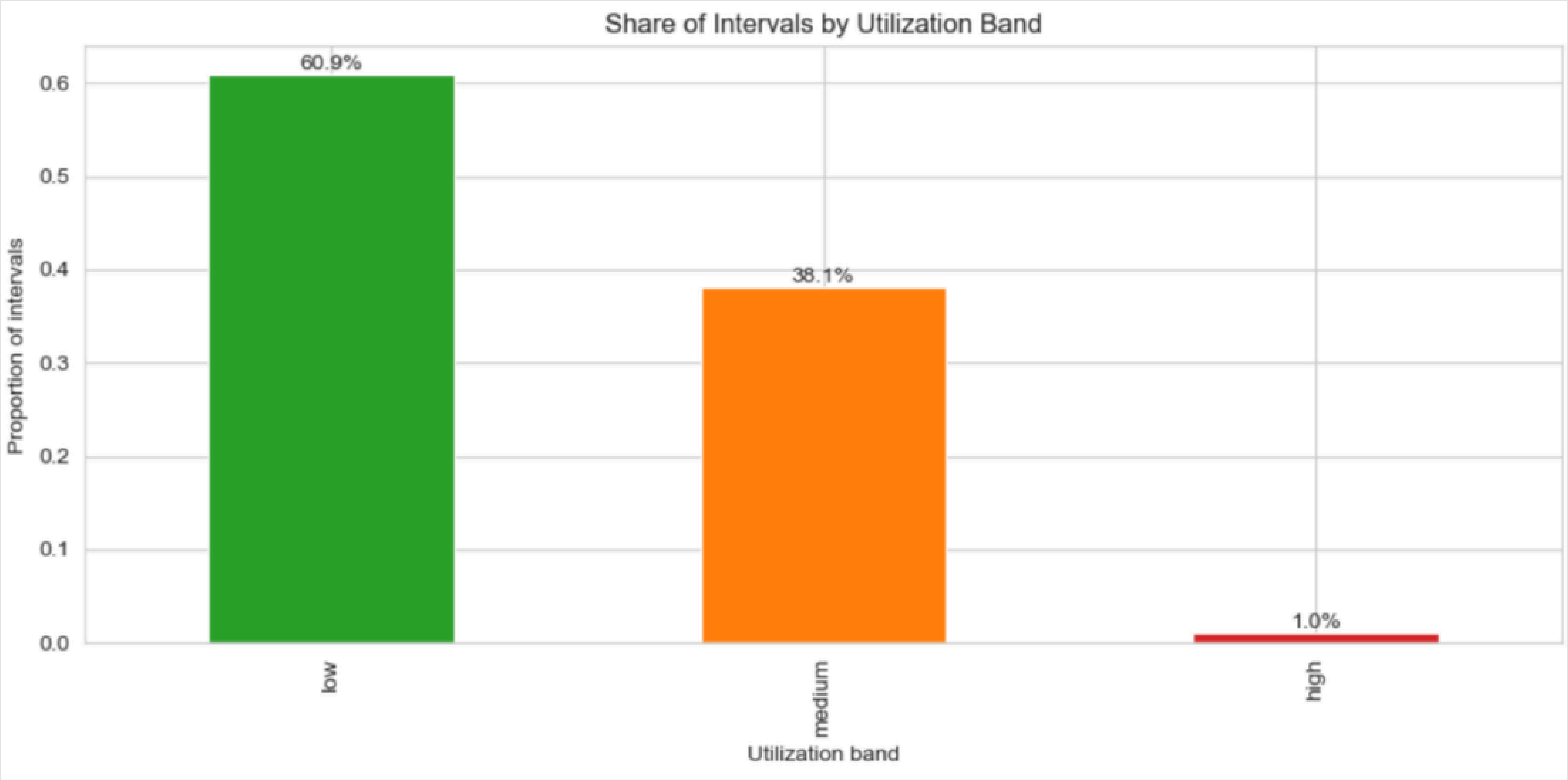
Reposition dynamic pricing as a grid-balancing tool, not a revenue lever

- **Business:** In low-utilization, price-inelastic networks, dynamic pricing won't grow revenue — but it eases congestion and improves utilization balance at near-zero cost
- **Operational:** Target discounts at price-sensitive non-CBD / fast-charger zones; conservative price bounds are sufficient and consumer-friendly
- **Policy:** Pricing can shift fleet charging off-peak (grid relief) without penalizing users — supporting managed EV growth and load-smoothing

Appendix A — Exploratory Analysis

- Inverted demand cycle — utilization peaks overnight, troughs midday, never nears the 80% surge line
- 61% of intervals fall below 30% utilization; only 1% exceed 80% — a predominantly under-utilized network





Appendix B — Demand Prediction Detail

- XGBoost selected — R^2 0.95, a 28% RMSE improvement over naive persistence

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Naive persistence   | RMSE: 0.0545 | R2: 0.9018
Linear              | RMSE: 0.0478 | MAE: 0.0287 | R2: 0.9245
RandomForest        | RMSE: 0.0423 | MAE: 0.0248 | R2: 0.9409
XGBoost             | RMSE: 0.0391 | MAE: 0.0236 | R2: 0.9495

Best utilization model: XGBoost
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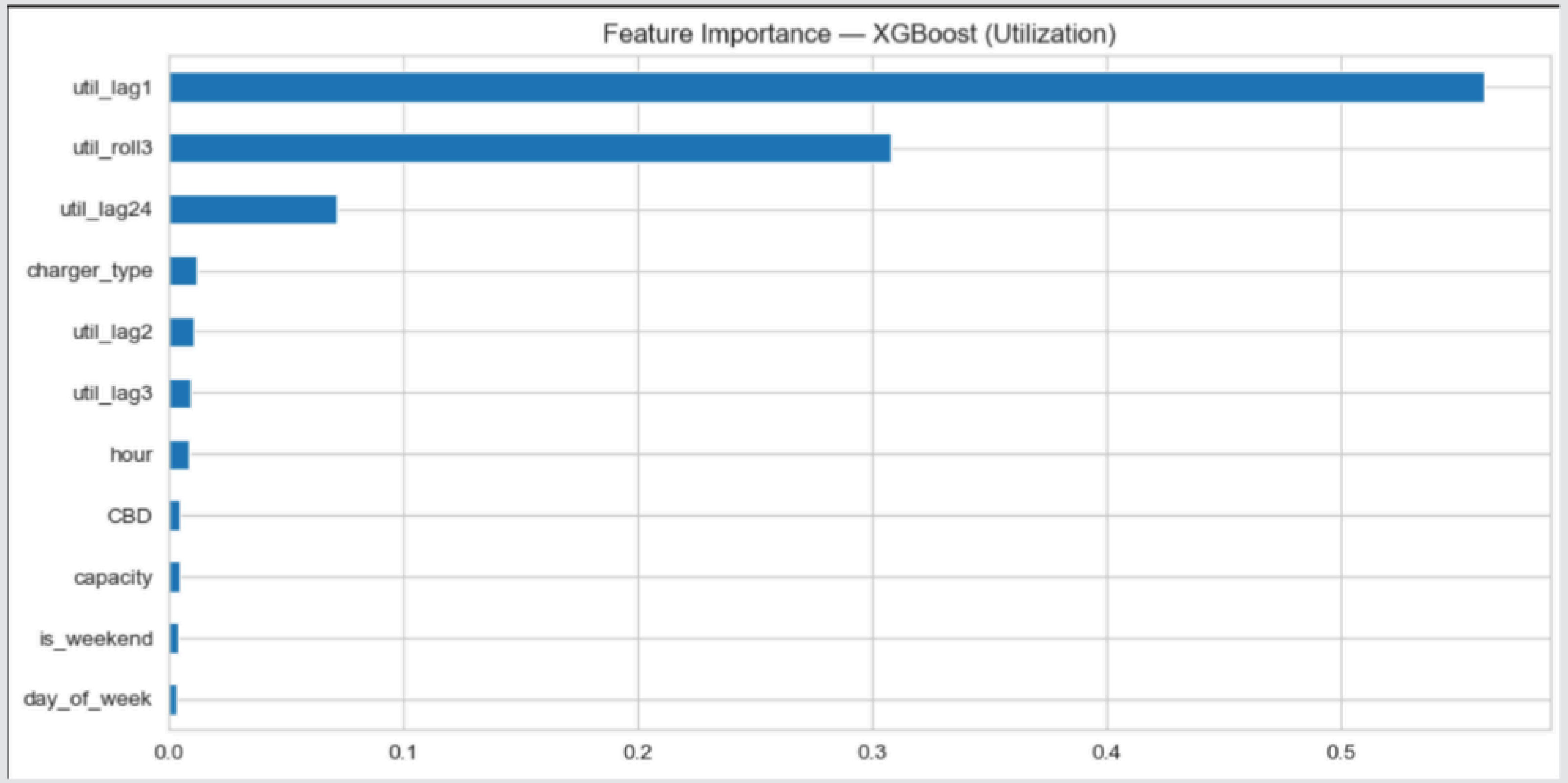
- Recent demand history dominates: previous-hour and 3-hour rolling mean drive most predictive power



Appendix C — Pricing Logic & Robustness

- Prices surcharge busy overnight hours (~₹15.5) and discount the quiet daytime (~₹14.5), mirroring demand

	hour	price_threshold	price_smooth	avg_util
0	0	15.44	15.50	0.34
1	1	15.42	15.52	0.35
2	2	15.43	15.48	0.34
3	3	15.41	15.47	0.34
4	4	15.38	15.45	0.34
5	5	15.37	15.46	0.34
6	6	15.42	15.50	0.34
7	7	15.04	15.26	0.32
8	8	14.52	14.91	0.28
9	9	14.20	14.76	0.26
10	10	13.85	14.64	0.25
11	11	13.76	14.58	0.24
12	12	14.09	14.72	0.26
13	13	14.22	14.77	0.26
14	14	13.94	14.65	0.25
15	15	13.78	14.61	0.24
16	16	13.66	14.53	0.24
17	17	13.71	14.54	0.24
18	18	13.84	14.61	0.25
19	19	14.32	14.86	0.27
20	20	14.53	15.00	0.29
21	21	14.71	15.06	0.30
22	22	14.81	15.15	0.31
23	23	15.09	15.27	0.32

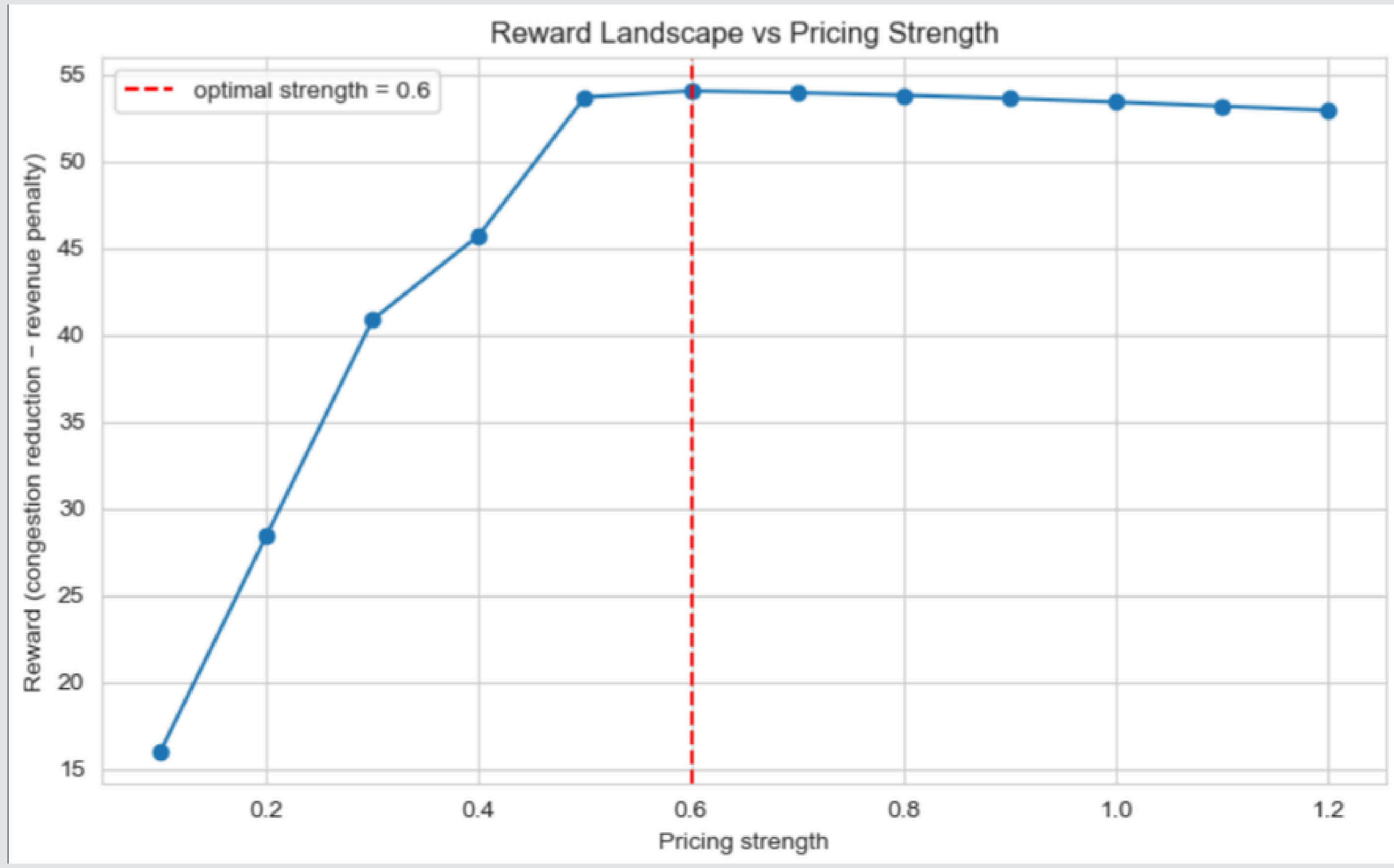
- Smooth strategy beats threshold at every elasticity; revenue stays near-neutral (-0.3% to -0.5%)

Baseline revenue (₹15 flat): 2025420.0

elasticity	strategy	revenue	revenue_gain_%	energy_kWh
-0.2	Threshold	1957994.0	-3.33	136125.0
-0.2	Smooth	2016326.0	-0.45	135175.0
-0.3	Threshold	1965432.0	-2.96	136674.0
-0.3	Smooth	2017285.0	-0.40	135248.0
-0.5	Threshold	1980308.0	-2.23	137771.0
-0.5	Smooth	2019202.0	-0.31	135395.0

Appendix D — Monitoring & Consolidated Scorecard

- Reward peaks at pricing strength 0.6 — a robust plateau; aggressive pricing adds no benefit



- Peak-shaving effect: peak-hour utilization falls more than the average, flattening the demand curve

measure	before	after	change_%
Peak-hour utilization	0.340	0.330	-2.8
Overall mean utilization	0.288	0.283	-1.7

- All three agents against all required metrics — strong forecasting, near-neutral revenue, 54% congestion relief

	agent	metric	result
1	Demand Prediction	Utilization RMSE	0.0391
2	Demand Prediction	Utilization MAE	0.0236
3	Demand Prediction	Utilization R2	0.95
4	Demand Prediction	Congestion ROC-AUC	0.997
5	Demand Prediction	Charging Load R2	0.974
6	Tariff Pricing	Gain % (smooth, e=-0.3)	-0.40%
7	Tariff Pricing	ain % (threshold, e=-0.3)	-2.96%
8	Tariff Pricing	ross elasticities (smooth)	-0.45% to -0.31%
9	Tariff Pricing	price bounds (±10–50%)	-0.84% to -0.38%
10	Monitoring & Learning	g Time Reduction (proxy)	54.46%
11	Monitoring & Learning	Customer Response Rate	+0.16%
12	Monitoring & Learning	Off-Peak Uplift	+0.45%
13	Monitoring & Learning	Pricing Efficiency (₹/kWh)	14.92 vs 15.00
14	Monitoring & Learning	pricing strength (learned)	0.60